

Fall Protection and Working at Heights Safety Standards

Reference: 29 CFR 1926.502 / 1910.23 - OSHA Fall Protection

Fall Hazard Statistics

Falls are the leading cause of death in construction, accounting for over 350 fatalities annually. In general industry, falls cause over 200,000 injuries per year. OSHA requires fall protection at 4 feet (general industry), 5 feet (maritime), and 6 feet (construction). Fall protection is required near any unprotected edge or opening.

Fall Protection Systems

Guardrail systems: top rail 42 inches (± 3), mid-rail at 21 inches, toe board 3.5 inches. Withstand 200 lb force. Safety net systems: installed within 30 feet below work area, withstand 17,500 lb drop test. Personal Fall Arrest Systems (PFAS): harness, lanyard, anchorage rated at 5,000 lb per worker.

Personal Fall Arrest System (PFAS)

Full-body harness must be worn (body belts are prohibited for fall arrest). Self-retracting lanyards limit free fall to 2 feet. Shock-absorbing lanyards allow up to 6-foot free fall. Anchorage points must support 5,000 lb. Perform pre-use inspection: check webbing for cuts, burns, chemical damage, and hardware for corrosion.

Ladder Safety

Portable ladders must extend 3 feet above landing. Set at 4:1 ratio (1 foot out for every 4 feet up). Never use top two rungs of a step ladder. Three points of contact at all times. Secure top and bottom to prevent movement. Rated for the combined weight of worker, tools, and materials. Inspect before each use.

Scaffolding Safety (29 CFR 1926.451)

Scaffolds must support 4x the maximum intended load. Planks must overlap 12 inches and extend 6-12 inches beyond supports. Guardrails required when scaffold is 10 feet or higher. Scaffold must be erected/moved/dismantled by a competent person. Access ladders or stairways required when scaffold is more than 2 feet above/below access point.

Fall Protection Training

Employees must be trained to recognize fall hazards and procedures to minimize them. Training topics: nature of fall hazards, correct use and operation of guardrail systems, PFAS, safety nets, covers, and warning lines. Retraining required when behavior indicates inadequate understanding or workplace changes create new hazards.

